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Experiment No. 3

Verification of Superposition Principle

Objective

The aim of this experiment is to verify experimentally the Superposition theorem which is an analytical technique of determining currents/voltages in a circuit with more than one emf source.

Theory

The Superposition Principle is a fundamental concept in electrical circuits that states that in any linear, active, bilateral network having more than one source, the response across any element is the sum of the responses obtained from each source considered separately and all other sources are replaced by their internal resistance. The superposition theorem is used to solve the network where two or more sources are present and connected. The current or voltage through any component in a circuit is the sum of the effects of each individual source acting alone. In other words, the principle states that **the total response of a circuit with multiple sources is the sum of the responses of the circuit to each individual source acting alone**. This principle is widely used in circuit analysis to simplify complex circuits and solve them with ease.

In a **linear circuit** containing multiple independent sources and linear elements (e.g., resistors, inductors, capacitors), the voltage across (or the current through) any element when all the sources are acting simultaneously may be obtained by adding algebraically all the individual voltages (or the currents) caused by each independent source acting alone, with all other sources deactivated.

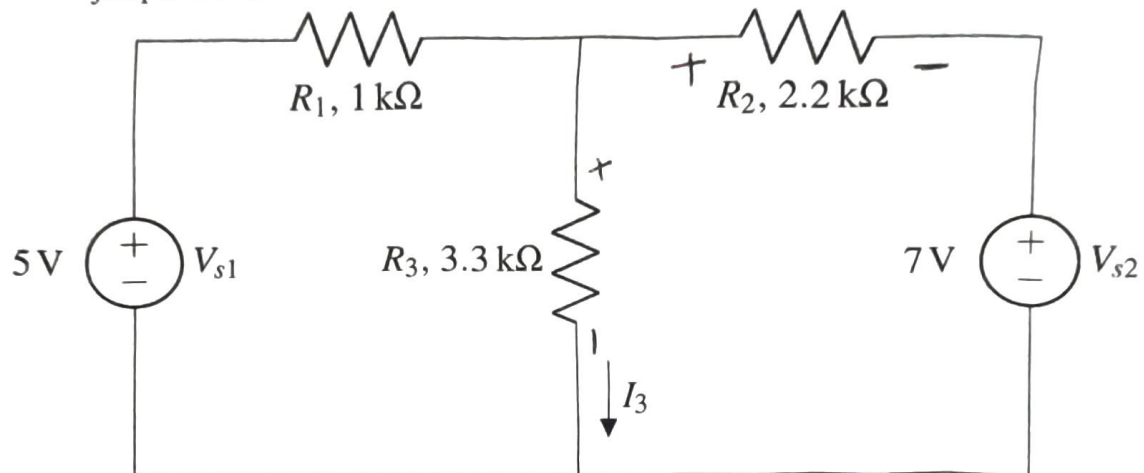
An independent **voltage source** is deactivated (made zero) by **shorting** it and an independent **current source** is deactivated (made zero) by **open circuiting** it. However, if a dependent source is present, it must remain active during the superposition process.

Apparatus

- Multimeter
- Resistors (1k, 2.2k, 3.3k).
- DC power supply
- Breadboard
- Jumper wires

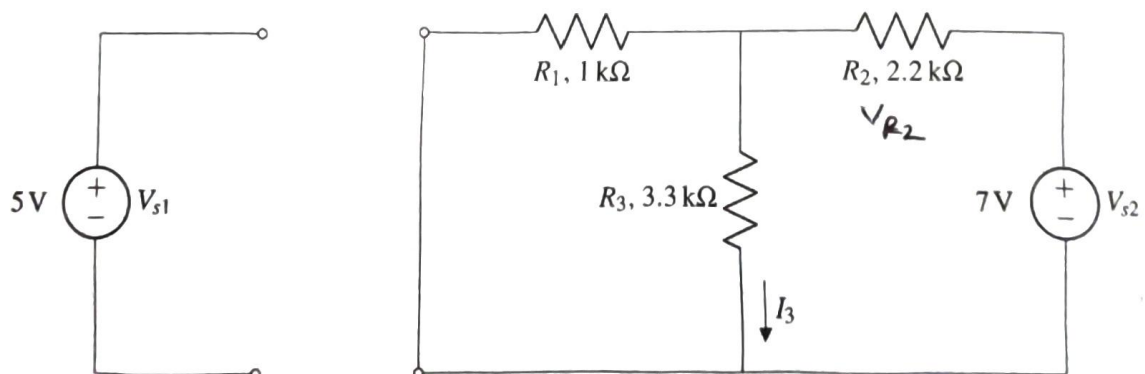
Procedures

- Measure the resistances of the provided resistors and fill up the data table (Table 1).
- Construct the following circuit on a breadboard. Try to use minimum number of jumper wires:



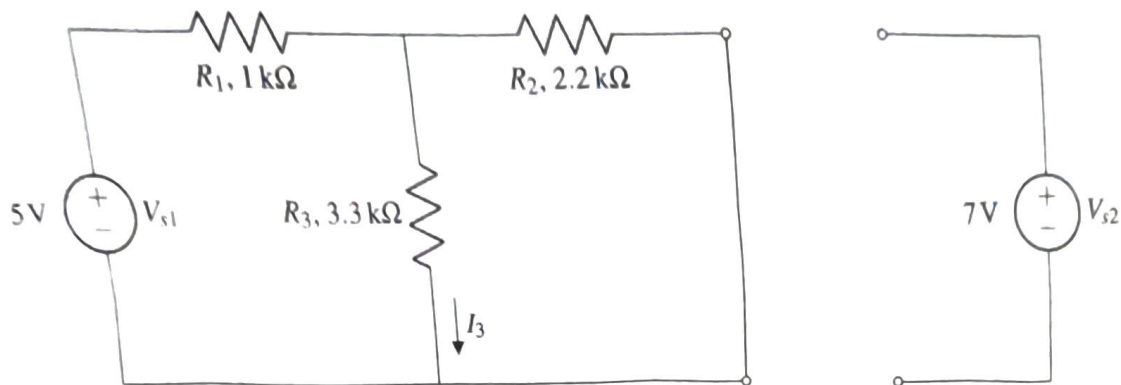
Circuit 1

- Measure the voltage across the resistors R_2 , R_3 and current through the resistor R_3 . Use a Multimeter for measuring the voltage and use Ohm's law to calculate the current (I_3) through R_3 . Fill up the data tables.
- Render V_{s1} inactive (keeping V_{s2} active) & Construct the following circuit.



Circuit 2

- Measure the voltage across the resistors R_2 , R_3 and current through the resistor R_3 . Use a Multimeter for measuring the voltage and use Ohm's law to calculate the current (I_3') through R_3 . Fill up the data tables.
- Render V_{s2} inactive (keeping V_{s1} active) & Construct the following circuit.



Circuit 3

- Measure the voltage across the resistors R_2 , R_3 and current through the resistor R_3 . Use a Multimeter for measuring the voltage and use Ohm's law to calculate the current (I_3) through R_3 . Fill up the data tables.
- Verify if $I_3 = I_3' + I_3''$ which would validate the superposition theorem for the current through R_3 .
- Verify if $V_2 = V_2' + V_2''$ which would validate the superposition theorem for the voltage across R_2 .

Data Tables

Signature of Lab Faculty:

S. Louis

Date:

15.6.23

**** For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

Table 1: Resistance Data

For all your future calculations, please use the observed values only (even for theoretical calculations).

Notation	Expected Resistance	Observed Resistance ($k\Omega$)
R_1	1 $k\Omega$	0.986
R_2	2.2 $k\Omega$	2.159
R_3	3.3 $k\Omega$	3.250

Table 2: Current through R_3

In the following table, V_3 is the voltage drop across resistor R_3 and I_3 is the current through it. Similar syntax applies to the remaining resistors. **For theoretical calculations, please note that, in series connection, the supplied voltage will be divided proportionally to the resistances.** Also, calculate the percentage of error between expected and observed values of $I_3' + I_3''$.

Observation	I_3 with both V_{s1} & V_{s2} active (mA)	I_3' with only V_{s1} active (mA)	I_3'' with only V_{s2} active (mA)	$I_3' + I_3''$ (mA)
Experimental	1.433 1.433	0.5563	0.8784	1.4347
Theoretical	1.409	0.5484	0.8606	1.409

• **Percentage of error** = $\left| \frac{\text{Observed Value} - \text{Expected Value}}{\text{Expected Value}} \right| \times 100\%$

N.B: Here, the Expected value is I_3 and the Observed value is $I_3' + I_3''$.

Hence, Percentage of error in $I_3' + I_3''$ calculation = 0.1186 %

Table 3: Voltage across R_2

In the following table, V_2 is the voltage drop across resistor R_2 . Similar syntax applies to the remaining resistors. **For theoretical calculations, please note that, in series connection, the supplied voltage will be divided proportionally to the resistances.** Also, calculate the percentage of error between expected and observed values of $V_2' + V_2''$.

Observation	V_2 with both V_{s1} & V_{s2} active (V)	V_2' with only V_{s1} active (V)	V_2'' with only V_{s2} active (V)	$V_2' + V_2''$ (V)
Experimental	-2.310	-5.17	2.854	-2.316
Theoretical	-2.35	-5.19	2.84	-2.35

Hence, Percentage of error in $V_2' + V_2''$ calculation = 0.259 %

Questions

1. Calculate the **power associated** with R_2 using the experimentally measured values of currents or voltages when:

- Only V_1 source is active.
- Only V_2 source is active.
- Both V_1 and V_2 sources are active.

Fill out the Table given below and verify, whether the superposition theorem is verified or not in this case. If not, comment on the reasons. **You don't need to take any new readings for this task.** Use previous data from Table 2 to calculate the power. Remember, power consumed by a resistor, $P = VI = I^2R = \frac{V^2}{R}$

Observation	P_{R_2} when both V_1 and V_2 active $P_{R_2} = \frac{V_{R_2}^2}{R_2}$ (W)	P_{R_2}' when only V_1 is active (7V) $P_{R_2}' = \frac{V_{R_2}^2}{R_2}$ (W)	P_{R_2}'' when only V_2 is active (5V) $P_{R_2}'' = \frac{V_{R_2}^2}{R_2}$ (W)	$P_{R_2}' + P_{R_2}''$ (W)
Experimental	2.4715	12.38	3.7727	16.1527
Theoretical	2.510	12.293	3.666	15.909

Is the Superposition Principle applicable in the case of Power?

☐ Yes

☒ No

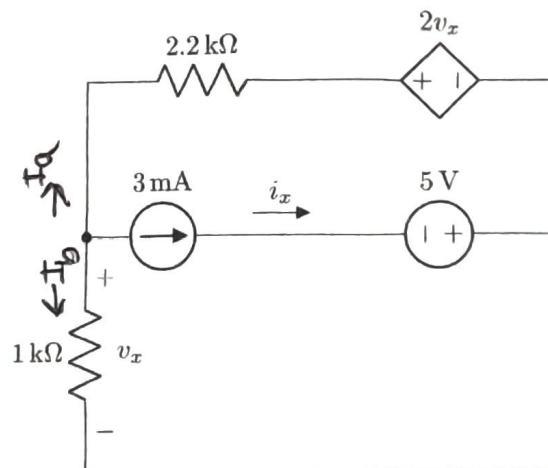
How would you relate your findings from this to the concept of linearity? Why does/doesn't it work when it comes to Power?

From the table we can see $P_{R_2} \neq P_{R_2}' + P_{R_2}''$
 So additivity is not present here.
 Also superposition principle won't work
 if the circuit is not linear. In this
 case the function of power $P = \frac{V^2}{R}$
 is not a linear function so superposition
 principle won't work in that case.

2. Why was a short circuit wire required to be connected between the corresponding terminals while turning off a voltage source? Why wasn't simply turning off the power switch enough to deactivate the source?

When a voltage source is connected to a circuit, it can store energy in various components like capacitors or inductors. Even after the power switch is turned off, these can retain a residual charge or energy. To ensure the isolation of voltage source we need to use jumper wire to short the circuit.

3. For the circuit shown below,



- (a) Show using the Superposition Principle that the 5 V voltage source has no effect on the current i_x .

From the diagram we can say that $i_x = 3 \text{ mA}$ and 5 V has no effect on i_x because i_x is an independent current source and an ideal independent source cannot be influenced by other thing. So, $i_x = 3 \text{ mA}$
 we can prove this through superposition principle by removing current source^{3 mA} on that branch and keeping its position open then

again removing 5V source with a short circuit.

For i_{x1} ,
As open circuit
 $i_{x1} = 0A$

For i_{x2} removing 5V source,

Applying KCL on the left node,

$$\frac{V_{x2} - 2V_{x2} + V_{x2} + 3 = 0 \Rightarrow 1.2V_{x2} = -6.6$$

$$\therefore V_{x2} = -5.5V \therefore I_a = 2.5mA, I_b = -5.5mA$$

We can verify,

$$I_a + I_b + i_{x2} = 0 \Rightarrow 2.5 - 5.5 + i_{x2} = 0$$

$$\therefore i_x = i_{x1} + i_{x2} = 0 + 3 = 3mA \therefore i_{x2} = 3mA$$

So, we can say 5V has no effect on i_x

I_a and I_b showed on the given circuit

(b) Why the 5V voltage source does not contribute to the current i_x ?

In the case of ideal current and voltage source if they are in series the no matter what the voltage source is the current source will always be dominant and that's why 5V source does not contribute to i_x .

(c) Can you draw any conclusions about the resistances of an ideal voltage and current source from this? If so, what are they?

An ideal constant current source has infinite resistance across it, so current would never flow through infinite resistance path because current always finds least resistance path. Hence total 3mA current flows through the straight path to 5V source.

And an ideal voltage source has zero resistance in series with it so, no restriction for current

Report

1. Fill up the theoretical parts of all the data tables.
2. Answer to the questions.
3. Discussion [comment on the obtained results and discrepancies]. Write in the next page.